Massachusetts Institute of Technology

GhostCAV

Distilling Interaction Preferences for Mixed-Autonomy Driving

Haopeng Deng

Mentor: Ryan Hardesty Lewis

PI: Prof. Cathy Wu

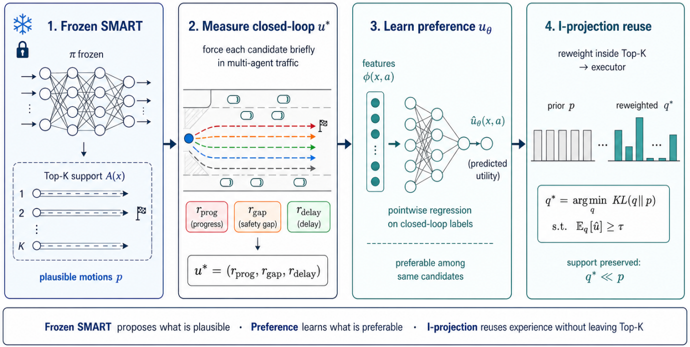
Summer Camp Midterm
August 3, 2026

Agenda

GhostCAV: Distilling Interaction Preferences for Mixed-Autonomy Driving

Haopeng Deng, Ryan Hardesty Lewis

Research question. Can a pretrained trajectory generator learn which plausible motion leads to a better interaction once other vehicles react?



Core innovation

Freeze SMART. Distill interaction preferences from closed-loop outcomes, then reuse them inside SMART's Top-K.

Preliminary finding

Held-out straight: 8.91 vs. 8.66 median efficiency (preference vs. random; $n=8$). **Directional only**; no intersection advantage yet.

Midterm progress

Platform construction
Complete
→
Method development
Complete
→
Experiments
In progress
→
Paper writing
In progress

01 | Motivation: reuse interaction traces

Can agents improve without direct communication?

Massachusetts Institute of Technology

Ants · pheromone

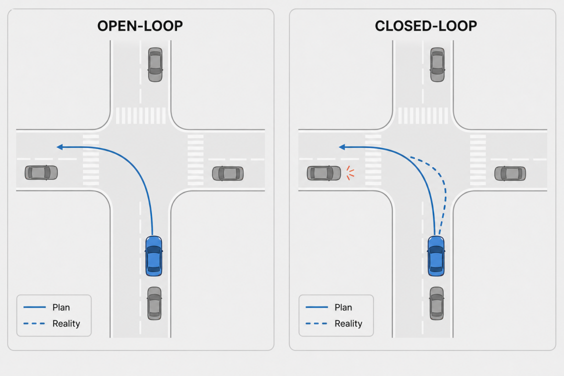
Ghost racing · past lap

Individuals do not talk. They leave **traces**; later agents **reuse** them.

GhostCAV applies this idea to closed-loop multi-agent driving.

01 | Motivation: Why closed loop matters

Massachusetts Institute of Technology



- Open-loop forecasts can look excellent on logs.
- In a long closed loop, other vehicles react; one choice changes the next interaction.
- We must run long enough to observe that downstream effect and turn it into a learning label.

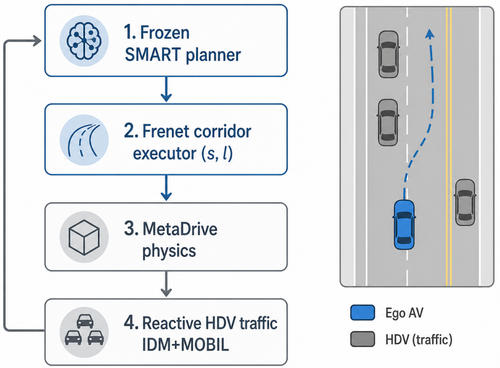
Our supervision is **multi-agent closed-loop outcomes**, **not a fixed recorded future**.

Why the rollout must be long enough

One choice changes the shared traffic state. The resulting merge, brake, or neighbour delay may appear seconds later; a sustained rollout reveals it as a reusable learning label.

02 | Platform: interaction-ready closed loop

SMART proposes; MetaDrive reacts in one shared world



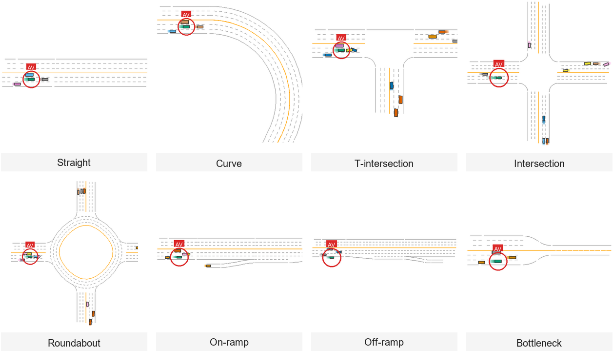
Refs: SMART—Zhou et al., NeurIPS 2024 MetaDrive—Li et al., TPAMI 2022

1. **SMART-7M**: lightweight autoregressive trajectory generator; CAT-K is the fine-tuned comparison.
2. It reads local map geometry and nearby agents' recent motion, then proposes Top-K waypoint futures.
3. **Frenet executor** applies (s, l, v): route progress / lateral lane position / target speed.
4. **MetaDrive** supplies physics; human-driven vehicles react through following and lane-change rules.

02 | Platform: scenarios, driver diversity, and networks

Massachusetts Institute of Technology

Train on Units, then compose them into road networks



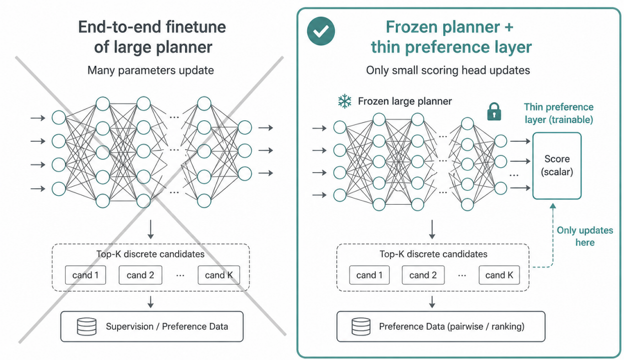
Eight reusable road families; the red ring marks the CAV.

MetaDrive—Li et al., TPAMI 2022   Scenario Factory and driver styles are designed in GhostCAV.

- **Scenario Factory** constructs Units with genuine interaction opportunities.
- Drivers vary from **cautious to aggressive**: speed, gap, and lane-change politeness.
- Units (junctions, merges, ramps, bottlenecks) join like building blocks into networks.
- **Soft topology routing** blends road-family experts at Unit boundaries rather than switching abruptly.

03 | Method: distilling interaction preferences

Past closed-loop outcomes train a preference that reweights SMART's Top-K



Refs: SMART—Zhou et al., NeurIPS 2024 CAT-K—Zhang et al., CVPR 2025

- Each past branch records safety, progress, and impact on nearby vehicles; these outcomes supervise u_θ.
- At a new observation, SMART proposes fresh Top-K futures; u_θ scores and reweights *these current candidates*.
- **CAT-K** changes SMART's weights toward logged trajectories; GhostCAV changes only current choice probabilities.

Pipeline: 1. Top-K $\to$ 2. measure $u^\star$ $\to$ 3. learn u_θ $\to$ 4. I-project $q^\star$

Massachusetts Institute of Technology

1. Top-K prior p

SMART scores next-token waypoints. Keep the eight highest as the action set

$$\mathcal{A}(x) = \{a_1, \dots, a_K\} \quad (K{=}8).$$

Softmax over those logits (temperature T) gives the renormalised prior on the menu:

$$p_a = \frac{e^{\ell_a/T}}{\sum_{b\in\mathcal{A}(x)} e^{\ell_b/T}}, \qquad a \in \mathcal{A}(x).$$

Later steps only redistribute mass inside $\mathcal{A}(x)$.

2. Closed-loop utilities $u^\star$

At sampled replans, force each $a \in \mathcal{A}(x)$ for a short closed-loop branch and record

$$\mathbf{u}^\star(x,a) = \left(r_{\text{prog}},\, r_{\text{gap}},\, r_{\text{delay}}\right):$$

path progress, minimum neighbour gap, and neighbour delay. Scalar utility for projection:

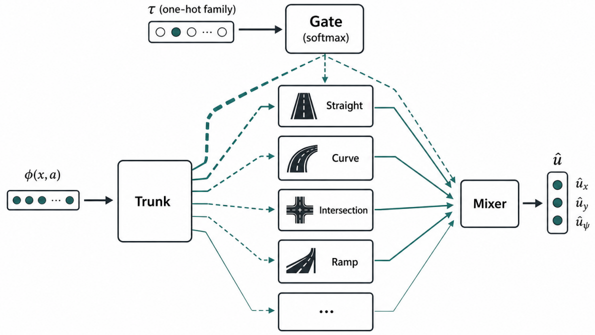
$$u(x,a) = w\cdot\mathbf{u}^\star(x,a).$$

Different w picks a behavioural style (e.g. more progress vs. more gap).

03 | Method: 3. learn u_θ

Soft topology mixture: share a trunk across road families, gate experts by Unit type τ

Massachusetts Institute of Technology



Predict the three heads of $\mathbf{u}^\star$ from ego-centric features $\phi(x, a)$. One expert per road family; soft gates mix them near Unit boundaries:

$$h = \mathrm{Trunk}(\phi)$$
$$g = \mathrm{softmax}(W e_\tau)$$
$$\hat{\mathbf{u}} = \textstyle\sum_i g_i\, e_i(h)$$

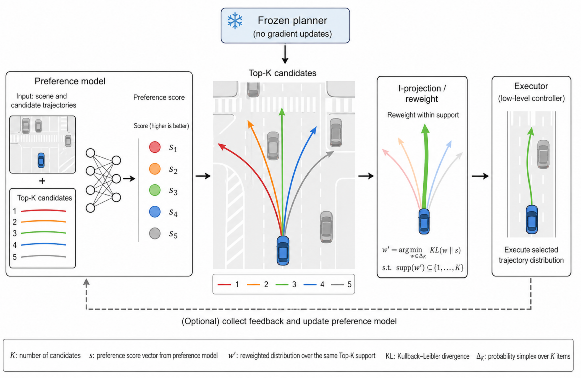
Pointwise MSE on standardised heads:

$$\min_\theta \; \mathbb{E}\big\|z_\theta(\phi) - z(\mathbf{u}^\star)\big\|_2^2.$$

At inference, $\hat{u} = w \cdot \hat{\mathbf{u}}$ enters step 4 only.

03 | Method: 4. I-project $q^\star$

Raise expected utility while staying close to the Top-K prior p



Closest q to p with a utility floor:

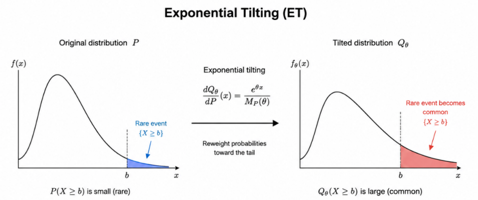
$$q^\star = \arg\min_{q \in \mathcal{C}} \mathrm{KL}(q \| p), \quad \mathcal{C} = \{q : \mathbb{E}_q[\hat{u}] \geq \tau(\kappa)\}$$

$$\tau(\kappa) = \mathbb{E}_p[\hat{u}] + \kappa(\max \hat{u} - \mathbb{E}_p[\hat{u}])$$

Closed form—tilt p by $\hat{u}$:

$$q_a^\star = \frac{p_a \, e^{\lambda^\star \hat{u}_a}}{Z(\lambda^\star)}$$

$\hat{u}$: predicted utility; $\lambda^\star$: dual for the floor; flat $\hat{u} \Rightarrow q^\star = p$.



Same mechanism on the Top-K menu: reweight p toward high $\hat{u}$.

04 | Evidence: protocol and held-out results

Three arms share the same Top-K menu; evaluate learned vs. random on held-out scenes

Massachusetts Institute of Technology

Protocol

Arm	Object	What it tests
Frozen	p	SMART alone
Learned	$q^{\star}(\hat{u})$	Distilled preference helps?
Random	$q^{\star}(u_{\text{rand}})$	Same operator, random scores

Intersection

	Eff.↑	Feas.↑	Gap↑	Coll.
Frozen	6.72	0.92	−0.40	0
Learned	**6.58**	**0.89**	**−0.44**	0
Random	6.73	0.93	−0.40	0

Straight

	Eff.↑	Feas.↑	Gap↑	Coll.
Frozen	8.05	0.92	1.60	0
Learned	**8.91**	**0.94**	**1.89**	0
Random	8.66	0.92	1.66	0

Reading

Intersection: data do **not establish** learned $>$ random. Straight: higher efficiency median than random (8.91 vs. 8.66), but $n{=}8$ is directional only.

Why the evidence is inconclusive

1. Interaction is still too weak. Junctions need yield/wait; current human drivers mostly follow and change lanes.
2. Limited samples. Sparse labels mean high variance. Next: thicken interaction realism, then expand held-out tests by topology—to limit overfitting.

05 | Next: Interaction Experience Bank + networks

Keep informative interaction scenarios, evict weak ones, relabel survivors, and retrain

Massachusetts Institute of Technology

Test learned vs. random

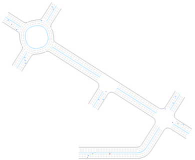
Keep useful / evict weak

Relabel + retrain u_θ

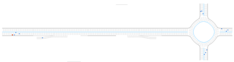
Re-test / composite roads

loop

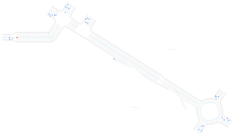
Fixed-capacity bank: keep informative scenes, evict weak ones—not trajectory replay at inference.



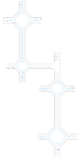
Urban mixed



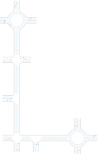
Highway mixed



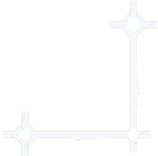
Topology tour



Multi-hub



Arterial mesh



Interchange

After Unit tests are stable: cross-topology generalization on composite networks.

Massachusetts
Institute of
Technology

Thank you

Questions welcome.